

IT353 BLOCKCHAIN TECHNOLOGY



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Module - V

- **Platforms and Applications:**

- Introduction to Blockchain platform:
 - Ethereum,
 - Hyperledger,
 - IOTA,
 - EOS,
 - Multichain,
 - SOLIDITY,
- Designing a new blockchain, Distributed Application (DAPP).

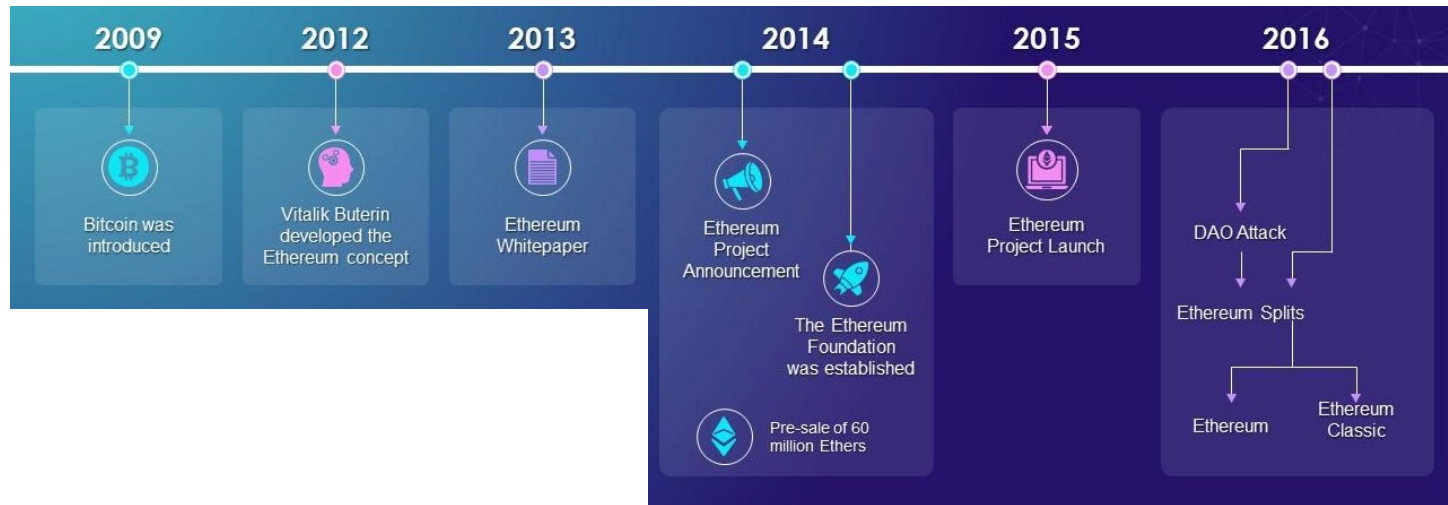
- **Applications:**

- Cryptocurrency
 - Healthcare
 - Finance and Banking
 - Real Estate
 - Retail
 - E-Governance & Voting
- Challenges and Research Issues in blockchain

Blockchain Platforms: Ethereum

Ethereum

- Ethereum is a world wide decentralized software platform based on blockchain technology.
- **Origin of Ethereum**



ethereum

<https://www.slideteam.net/the-origin-and-timelines-of-ethereum-ether-training-ppt.html#images-1>

Blockchain Platforms: Ethereum

- It offers programmability feature through smart contracts.
- It uses Ether (ETH) as its cryptocurrency.
- The main function of Ethereum is its programmability, hence ease of application building.
- Two types of Account: Externally Owned Accounts and Contract Accounts
- **Components of Ethereum**
 - ***State Machine***: Transactions and state changes are processed by a state machine called the Ethereum Virtual Machine (EVM).
 - ***P2P network***: All nodes are connected to each other
 - ***Nodes***: Any device connected to the blockchain
 - ***Transactions***: The message being exchanged on the network.
 - ***Consensus Algorithm***: Used to achieve mutual agreement

Blockchain Platforms: Ethereum

Terminologies

- **Gas**

- Gas is needed to run applications on the Ethereum network.
- Gas is a unit that measures the computational power required to run a smart contract or a transaction.
- The transaction fees equal the amount of gas required to execute a transaction multiplied by the gas price.
- “Gas limit” refers to the amount of gas used for the computation and the amount of ether a user is required to pay for the gas.

- **Ether (ETH)**

- It is Ethereum’s cryptocurrency.
- It is used to pay for the computational resources and the transaction fees for any transaction executed on the Ethereum network.
- It is also used to buy gas, which is used to pay for the computation of any transaction made on the Ethereum network.

Blockchain Platforms: Ethereum

Terminologies

• Smart Contracts

- A smart contract is a simple computer program that facilitates the exchange of any asset between two parties.
- A program that executes on Ethereum are called smart contracts
- It is a self executing contract with certain terms (rules) between the buyer and seller.
- It is immutable (cannot be changes after deployment). If a modification needs to be done in future, the transactions correlated with the original contract cannot be altered.

• EVM

- A runtime compiler to executes smart contracts.
- Programs are written in high level language (Solidity), compiled and then converted to bytecode.
- Once the code is deployed on the EVM, every participant on the network has a copy of the contract.
- If X wants to pay Y 5 ETH, the transaction will be sent to the EVM using a smart contract for a fund transfer from X to Y. To validate the transaction; the Ethereum network will perform consensus.

Blockchain Platforms: Ethereum

Terminologies

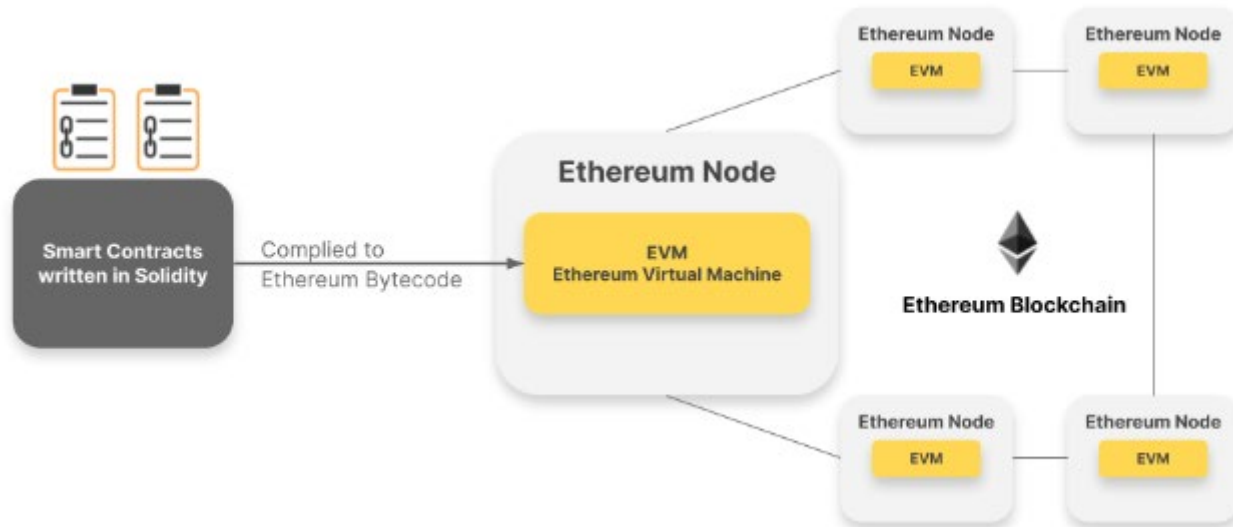
- **Solidity**

- It is a programming language created by Ethereum and released in the year 2015 and led by Christian Reitwiessner.
- It is a high-level programming language designed for implementing smart contracts.
- It is a statically typed object-oriented (contract-oriented) language.
- Solidity is highly influenced by Python, C++, and JavaScript which run on the Ethereum Virtual Machine (EVM).
- Solidity supports libraries, and inheritance.
- Solidity is the primary language for blockchains running platforms to create contracts.

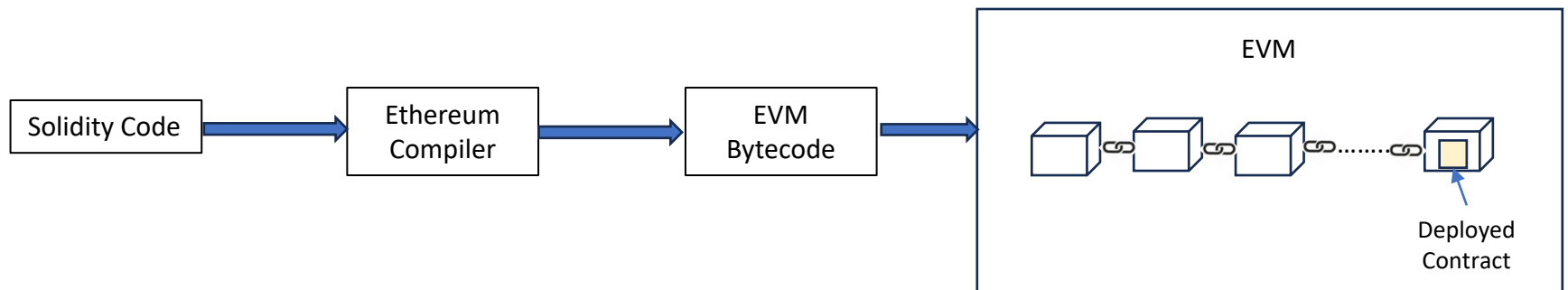
Blockchain Platforms: Ethereum

Terminologies

- EVM

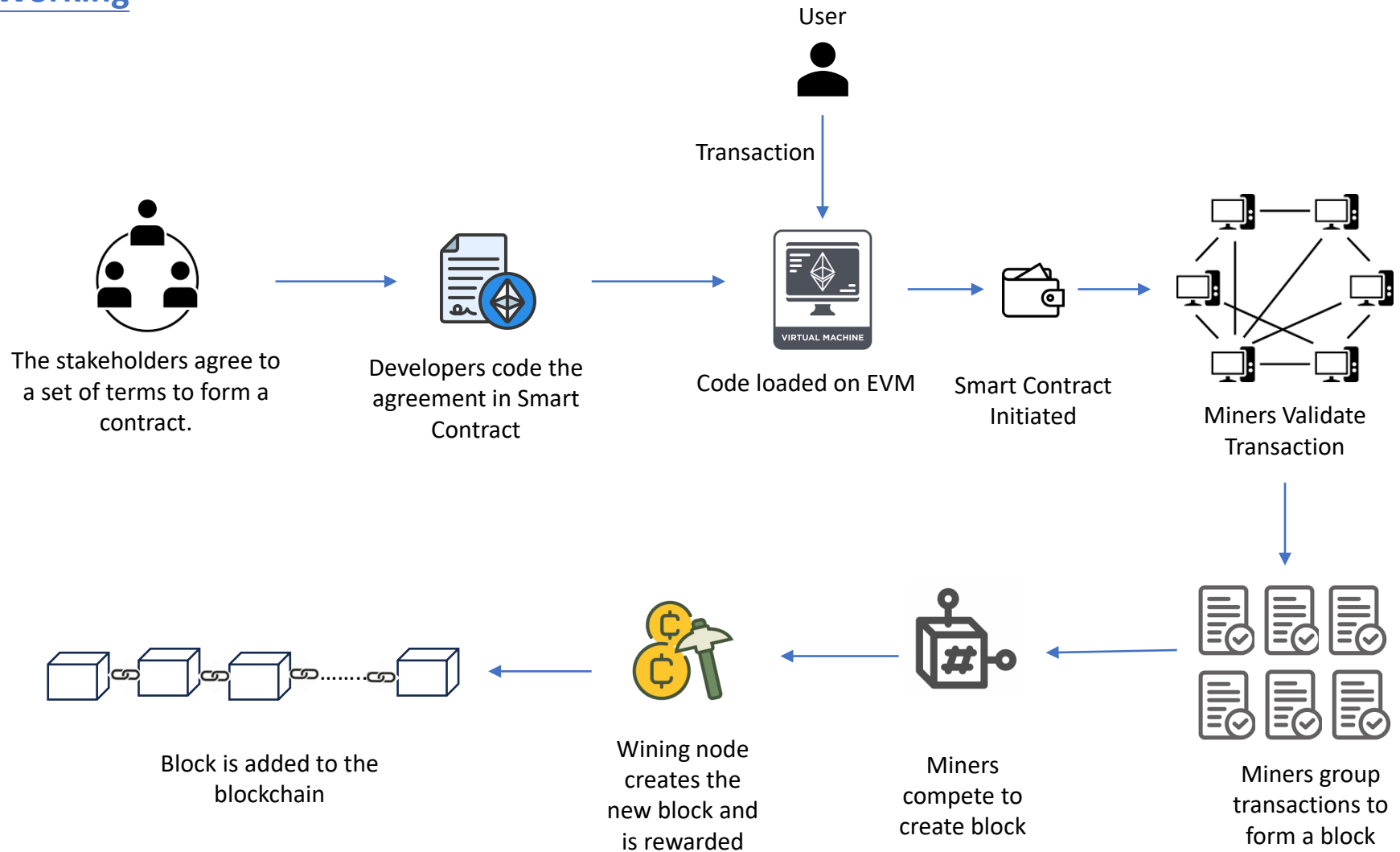


<https://coin98.net/what-is-evm>



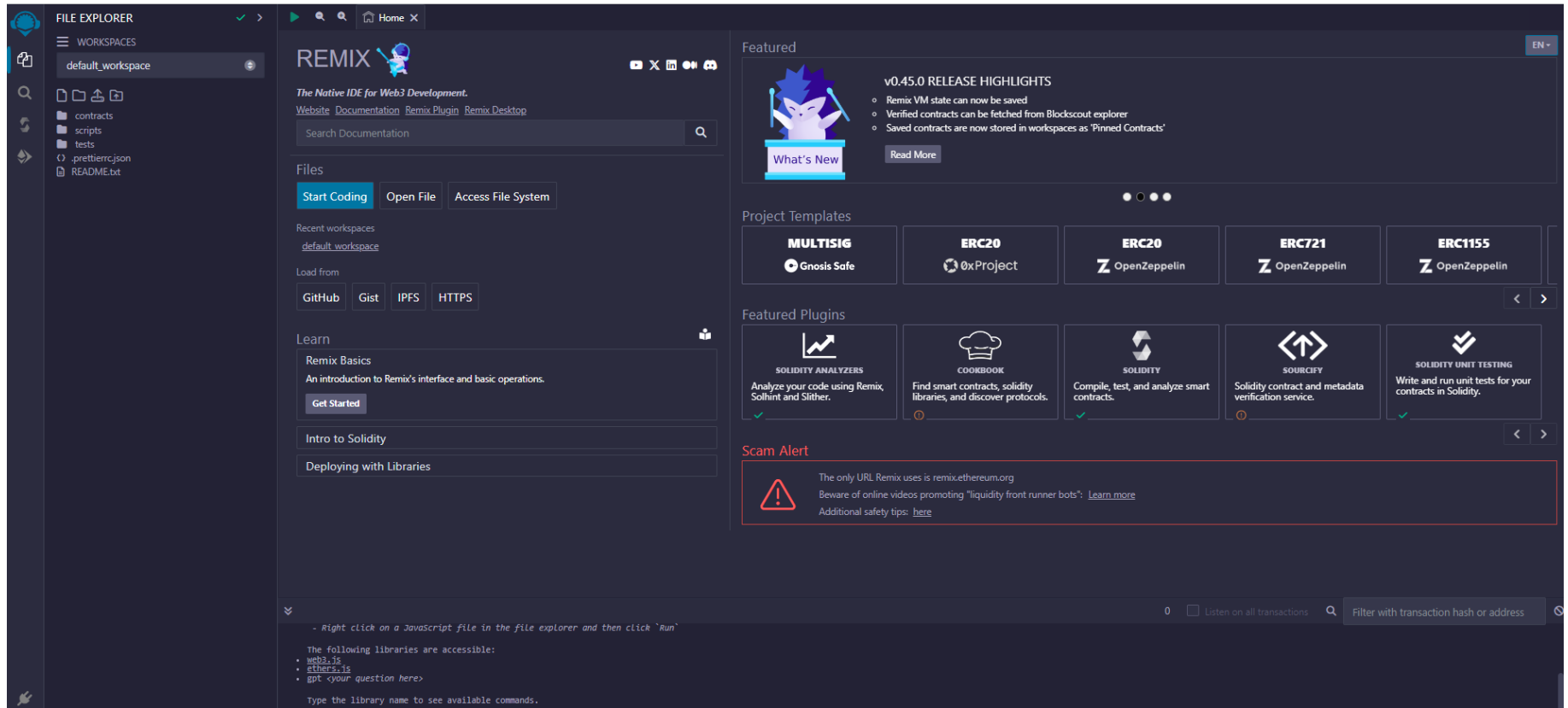
Blockchain Platforms: Ethereum

Working



Blockchain Platforms: Ethereum

Example



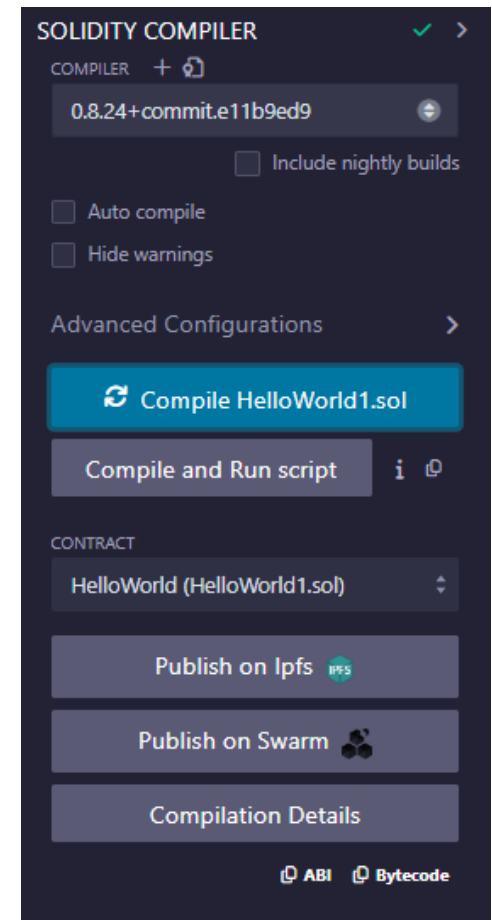
Remix IDE

Blockchain Platforms: Ethereum

Example

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity >=0.6.12 <0.9.0;
3
4 contract HelloWorld {
5     /**
6      * @dev Prints Hello World string
7      */
8     function print() public pure returns (string memory) { infinite gas
9         return "Hello World!";
10    }
11 }
12
```

Code the Smart Contract



Compile

Blockchain Platforms: Ethereum

Example

DEPLOY & RUN TRANSACTIONS ✓ >

ENVIRONMENT

Remix VM (Shanghai) ⓘ

VM

ACCOUNT ⓘ

0x5B3...eddC4 (100 ether) ⓘ ⓘ

GAS LIMIT

3000000

VALUE

0 Wei ⓘ

CONTRACT

HelloWorld - contracts/HelloWorld1.sol ⓘ

evm version: shanghai

Deploy

☐ Publish to IPFS

At Address Load contract from Address

Transactions recorded 0 ⓘ >

DEPLOY & RUN TRANSACTIONS ✓ >

ENVIRONMENT

Custom - External Http Provider ⓘ

Custom (5777) network

ACCOUNT ⓘ

0xF67...034aa (99.9934000722) ⓘ ⓘ

GAS LIMIT

3000000

VALUE

2 Ether ⓘ

CONTRACT

HelloWorld - contracts/HelloWorld.sol ⓘ

evm version: shanghai

Deploy

☐ Publish to IPFS

At Address Load contract from Address

Transactions recorded 2 ⓘ >

External Http Provider ✕

Note: To use Geth & <https://remix.ethereum.org>, configure it to allow requests from Remix(see [Geth Docs on rpc server](#))

geth --http --http.corsdomain <https://remix.ethereum.org>

To run Remix & a local Geth test node, use this command: (see [Geth Docs on Dev mode](#))

geth --http --http.corsdomain="<https://remix.ethereum.org>" --http.api web3,eth,debug,personal,net --vmdebug --datadir <path/to/local/folder/for/test/chain> --dev console

WARNING: It is not safe to use the --http.corsdomain flag with a wildcard: --http.corsdomain *

For more info: [Remix Docs on External HTTP Provider](#)

External HTTP Provider Endpoint

<http://127.0.0.1:7545>

OK Cancel

Connect to test network on Ganache

Deploy

Blockchain Platforms: Ethereum

Example

The screenshot displays the Ganache application window. The top navigation bar includes icons and labels for ACCOUNTS, BLOCKS, TRANSACTIONS (highlighted), CONTRACTS, EVENTS, and LOGS. A search bar on the right prompts the user to 'SEARCH FOR BLOCK NUMBERS OR TX HASHES'. Below the navigation bar, a status bar provides key network information: CURRENT BLOCK 2, GAS PRICE 20000000000, GAS LIMIT 6721975, HARDFORK MERGE, NETWORK ID 5777, RPC SERVER HTTP://127.0.0.1:7545, MINING STATUS AUTOMINING, and a WORKSPACE QUICKSTART button. To the right of this bar are buttons for SAVE, SWITCH, and a settings gear icon.

The main content area shows two transaction details. Each entry includes a TX HASH, a FROM ADDRESS, a CREATED CONTRACT ADDRESS, GAS USED, and VALUE. A red 'CONTRACT CREATION' button is positioned to the right of each transaction's hash.

TX HASH	FROM ADDRESS	CREATED CONTRACT ADDRESS	GAS USED	VALUE
0x139d56f5d73e98f8d033450508eae0fd091d0adf938e73fe0a7b1cdc3ec2ff6f	0xF675fBA0924A95D6B30d351CcFe6ec95428034aa	0x4bCE5e26159E6866f72eab1230f81086e0E67C28	3000000	2000000000000000000
0xccdd70454c929c9eee9a6c07c0429e1b6b5147410a281040c7263a1c03926998	0xF675fBA0924A95D6B30d351CcFe6ec95428034aa	0x4eE5e8C8FA1303dab98D8d2Cc06dFcb1CAA746b6	3000000	0

Blockchain Platforms: Hyperledger

Hyperledger

- Hyperledger is a collection of open-source projects created to support the development of blockchain-based distributed ledgers.
- Hyperledger provides the platform to create personalized blockchain services according to the need of business work.
- Hyperledger vs. Ethereum

Hyperledger	Ethereum
Used to build permissioned network	Used to create public Network
Chaincodes	Smart Contracts
Go language	Solidity language
No consensus needed	PoW, PoS
Faster	Slower
No native cryptocurrency	Ether

Blockchain Platforms: Hyperledger

Hyperledger

- Hyperledger-based technology works using these layers:
 - A consensus layer, which makes an agreement on order and confirms if the transactions in a block are correct.
 - A smart contract layer, which processes and authorizes transaction requests
 - A communication layer, which manages peer-to-peer (P2P) message transport.
 - An API, which allows other applications to communicate with the blockchain.
 - Identity management services, which validates the identities of users and systems.



HYPERLEDGER

Blockchain Platforms: IOTA

IOTA

- IOTA is an open-source distributed ledger and cryptocurrency designed for the Internet of things.
- It uses a directed acyclic graph to store transactions on its ledger.
- IOTA does not use miners to validate transactions, instead, nodes that issue a new transaction on the network must approve two previous transactions.
- Transactions can therefore be issued without fees and the network currently achieves consensus through a coordinator node, operated by the IOTA Foundation.
- The Tangle is IOTA's network. It immutably records the exchange of data and value. It ensures that the information is trustworthy and cannot be tampered with nor destroyed.



Blockchain Platforms: EOS

EOS

- EOS blockchain ecosystem is an open-source blockchain platform that aims to solve scalability issues for decentralized applications.
- EOSIO is an open-source blockchain software platform for developing scalable blockchain apps and networks.
- It is like an operating system which allows programmers to use Web Assembly languages like C++, Java, and Python to build blockchain apps.
- The EOSIO software runs the blockchain based on a Delegated Proof-of-Stake (DPoS) consensus mechanism for enhanced scalability.
- In 2017, the private company, Block.one published the EOS white paper.



Multichain

- Multi-chain is a process where projects deploy smart contracts across multiple blockchains, connecting isolated chains together as one network.
- The result is users can transact across many chains without needing to shuffle assets from one place to another.
- Enables a user to access an application on another blockchain.
- In a multi-chain application, developers configure a separate and isolated instance of their smart contracts for each different blockchain.
- A user who is already onboarded onto a different blockchain must take many steps to access a new application—for example, adding a new network or downloading a new wallet, bridging funds from one blockchain to another, and then acquiring the native gas token to transact.
- Benefits of Multi-chain
 - Scalability
 - Parallel Processing
 - Interoperability

Disturbed Applications (DApps)

- DApps are applications or software that run on multiple computers within a network at the same time and can be stored on servers or cloud computing platforms.
- Unlike traditional applications that run on a single system, distributed applications run on multiple systems simultaneously.
- DApps run on distributed computing systems, which are a collection of independent computers that appear to the user as a single system.
- They are more resistant to certain cyber attacks because they have no single point of failure.
- They can communicate with multiple servers or devices on the same network from any geographical location. The distributed nature of the applications refers to data being spread out over more than one computer in a network.
- Distributed applications are broken up into two separate models -- the client software and the server software. The client software or computer accesses the data from the server or cloud environment, while the server or cloud processes the data.

Applications of Blockchain

Cryptocurrency:

- No geographical limitations
- Different exchange rates



Healthcare

- Provides a decentralized system to store and manage EHRs securely and provides a single view of a patient's record.
- It also gives patients the freedom to approve changes to their EHRs, authorize who can view them, and control how they are shared by healthcare providers.
- Blockchain serves as the single source of truth for research and clinical trial data. It improves and secures record keeping and sharing while maintaining patient privacy.



Applications of Blockchain

Finance and Banking

- Money transfers using blockchain can be less expensive and faster than using existing money transfer services.
- Cross-border transactions
- Lenders can use blockchain to execute loan management through smart contracts to automatically trigger things like a service payment, a margin call, full repayment of the loan, and release of collateral.



Real Estate

- Asset tokenization platforms built on blockchain let you create a digital token of ownership for real-world liquid assets. Once assets are tokenized, investors only need a trading app to execute borderless international trading.
- Transparency in ownership rights and loan payment.
- Land registration and property management.



Applications of Blockchain

Retail

- Inventory Management
- Supply Chain Management
- Transparent product reviews



E-Governance and Voting

- Using blockchain technology can make sure that nobody votes twice, only eligible voters are able to vote, and votes cannot be tampered with.
- Administration of government benefits such as welfare programs. Using blockchain technology could reduce fraud and the costs of operations. Meanwhile, beneficiaries can receive funds more quickly through digital disbursement on the blockchain.



Challenges in Blockchain

- **Scalability**

- Blockchain networks can be slow and inefficient due to the high computational requirements needed to validate transactions. As the number of users, transactions, and applications increases, the ability of blockchain networks to process and validate them in a timely way becomes strained.

- **Energy Consumption**

- The process of validating transactions on a blockchain network requires a lot of computing power, which in turn requires a lot of energy. This has led to concerns about carbon emissions and the environmental impact of blockchain technology. Some blockchain projects have adopted alternative consensus mechanisms, such as PoS, which consume significantly less energy.

- **Complexity**

- Blockchain is a complex technology that requires a high level of technical expertise to implement and maintain. Efforts to address this issue include the development of user-friendly interfaces, streamlined onboarding processes, and educational resources that simplify the complexities of blockchain.

- **Interoperability**

- Ability of different blockchain networks to communicate and interact with each other
- This lack of interoperability can lead to inefficiencies, as individuals and companies may need to navigate multiple platforms and use a number of tokens or cryptocurrencies to interact with different networks.